

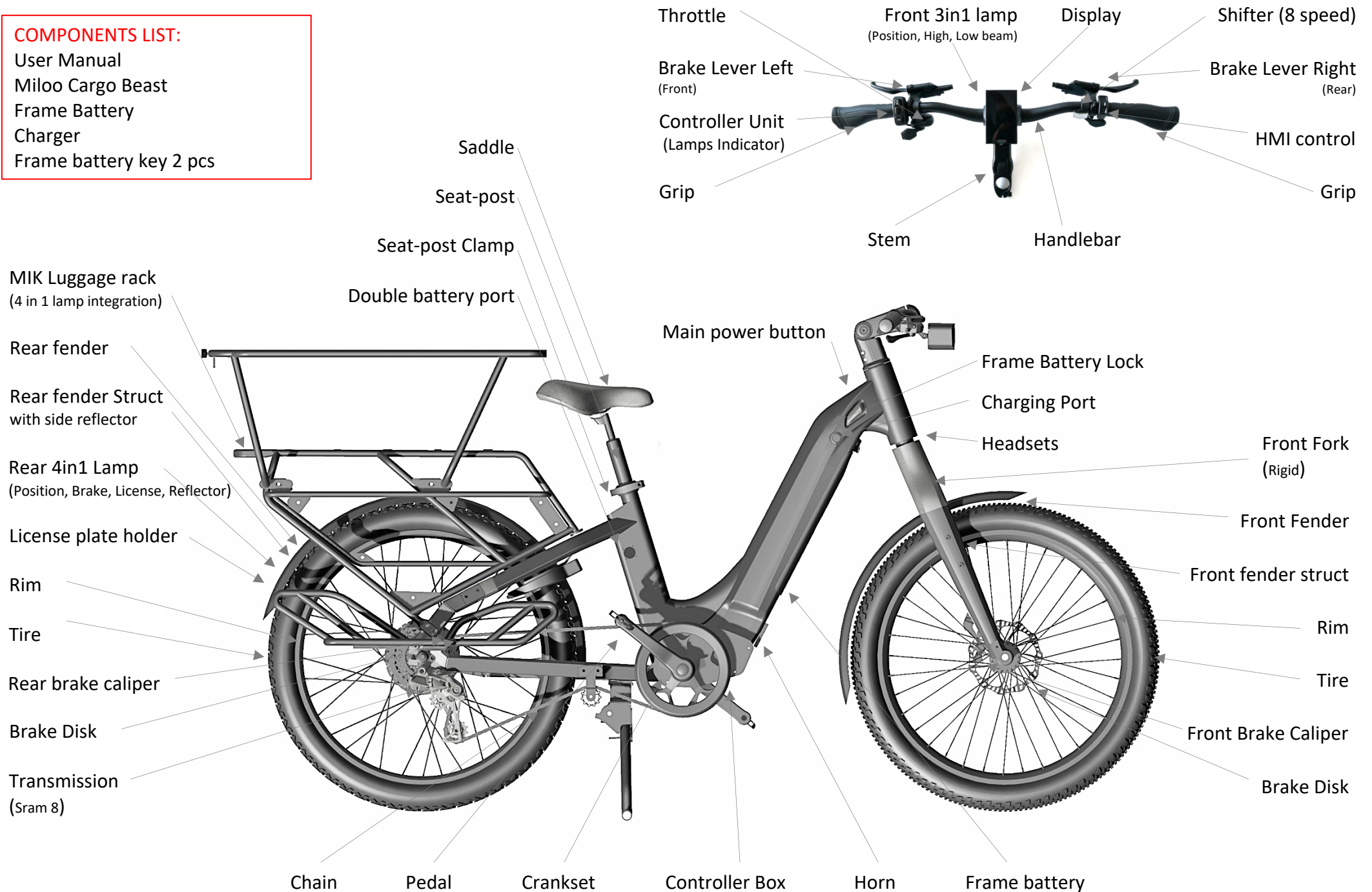


CARGO BEAST

User Manual | English

COMPONENTS LIST:

User Manual
Miloo Cargo Beast
Frame Battery
Charger
Frame battery key 2 pcs



Dear Customer

First of all, on behalf of the whole Miloo team, we would like to congratulate you on your purchase of your Cargo Beast.

Being part of change is no joke, so we thought it would be beneficial for you if we gave you some important information regarding your brand-new beast. This will help you use this incredible piece of engineering without any problems for a smoother and safer ride. This information is essential so please be careful when reading this user manual and keep it in a safe place.

Upon delivery, your bicycle has been assembled and adjusted to your specific needs. If this isn't the case, please contact our local partners so that they can take care of it for you.

We also hope that you understand the basic knowledge to drive this beast. Anyone involved in: using, repairing, maintaining or cleaning, or disposing of the bike must have read and understood the entirety of this user manual. Make sure to contact us if you have any additional questions or if you do not understand some of the information provided in this booklet. Your safety is our number one concern.

All information included in this user manual relates to the good use, structure, technology, care and maintenance of the beast you have received. Please follow all information as it is often related to your safety and ignoring it might lead to unfortunate accidents that could have been avoided. We don't want to bore you with useless information, so everything included is only the essential needed to use such a powerful piece of technology.

The instructions of the respective manufacturers of the components used must be consulted for special, technical details. They can be found on our website.

For more information, please contact us at info@miloo.com or call +41 79 158 62 91

i Verify that your driving license and personal protective appliances are compatible with the beast you have purchased before you first ride it.

i These user instructions are part of the Miloo Beast delivery scope and only applies to this model. Your Own Beast may look different.

For your safety

Please read through all the warnings and instructions in this user manual before riding your new Beast. Please make sure that you always store this booklet near your bike so that it may be accessible whenever it is needed.

Before enjoying your Beast for the first time, please go through the “Before the first ride” and “Before each ride”! It may contain information that could be useful for you.

Before letting your friends or family use your beast, make sure to give them this booklet as well in case they also need. These instructions include symbols and information that will require your outmost attention. When you see one of these symbols, the information that follows could be crucial as the problem could very well come to pass. The text that relates to this warning is clearly identified as red text.



Modern pedelec technology is very sophisticated. Working on it requires technical knowledge, experience and special tools. Do not work on your beast alone. Take your s-pedelec to a specialist workshop when it is in need of repair, maintenance or restoration.

Regularly check that all screw connections are appropriately tightened, and all components are not damaged.

i Note: This symbol informs you about handling the product, or the part covered in the user manual and should be read with care.



Warning: This symbol provides you with information that could prevent the misuse of the product which could result in damaging the beast or the environment.



Danger: This symbol warns of a possible risk of injury when the appropriate, required actions are not implemented or the respective precautionary measures are not taken.

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Quick Start guide for your Beast

! Here is a quick list of instructions on how to quickly get started with your new Cargo Beast. To maximize your safety, please ensure that you have fully read the entire manual, even for experienced S-Pedelec riders. More information can also be found online or by contacting us.

1. Charge completely the batteries before your first ride. You may do this in two different ways.

You can separately charge the battery from the bike by removing.

Frame battery: - Insert the key in its slot on the left-hand side of the main frame of the Beast and turn it anticlockwise

- Turn the battery switch below the main frame
- Pull the battery out
- Peel off the rubber seal from the socket placed at the bottom of the battery and plug it in
- After putting back the seal, slide the battery back in pins facing downwards
- Lock it back into place by turning the key clockwise

You can also directly charge your S-Pedelec by peeling off the rubber seal from the socket placed on the right-hand side of the main frame and plugging it in.

! The beast will not power on if the beast is plugged to ensure your safety and to preserve the charger.



Only use the charger available delivered with the bike to recharge the battery.

2. Power on the electrical system to use the electric motor's support:

- Press the power button placed on the right hand side of the bike's main frame, till a white light is emitted.
- Press the power button on the HMI controller positioned on the handlebar left for more than 2 seconds for it to activate.



The electric motor will only engage once you have already pedaled or by pressing on the throttle. Ensure that you practice riding the beast on safe and quiet roads, the powerful acceleration can surprise. Uncontrolled starts in public roads in traffic can lead to severe injuries.



Always make sure that your batteries are in place when riding. Some electric components will be in the open and could go through possible water damage. This could lead to electrical failure of the beast and possible injuries.

i The Cargo Beast can be driven without any electrical support if chosen.

The amount of the motor's support will depend on the amount chosen on the display; pedaling at any cadence will lead to the same speed. The motor will stop if you stop pedaling or let go of the throttle and will only come back once you start pedaling or press on the throttle again. Braking will also stop the motor thanks to the integrated sensors.

i Changing the support level can be done at any time during the ride safely without needing to take your hands off the handle by pressing on the buttons of HMI control.

3. To turn off the electrical system after every ride, press the power button on the side of the bike frame until it no longer emits any light.



If you only turn off the display on the handlebar, the front light and back light will remain and so continue depleting the battery around 10 minutes, so make sure that you have fully deactivated the Beast after every ride.

Safety instructions

Before riding your s-pedelec on public roads, you need to inform yourself of your own country's national regulation. For example, in many countries it is necessary to have a driving license in order to use a S-Pedelec beast.

This user manual also assumes that you, the rider, know how to ride a bicycle. This manual is not intended to teach you how to ride an Pedelec, neither is it intended to inform you on the assembly or repair of the beast.

Please be aware that this bike still contains more risk than other types of bikes. You are not protected by an external body, nor do you have any protections in case of a fall. Even though you are smaller than other beasts and drive on different parts of the road than pedestrians, you are also faster than what might be anticipated so please pay attention to others on the road.

Here is the list of actions that need to be considered while using your new Beast:

- **Always wear a dedicated S-Pedelec helmet for every ride.**
- Make sure that you thoroughly understand the safety instructions for your specific helmet.
- When driving, bright and reflective clothing is recommended in order to be seen by others.
- Baggy and loose clothing is also not recommended, and the soles of your shoes should also have non-slip characteristics.
- **Always keep both hands on the handlebar.**
- Keep your ears clear from headphones or other devices of the sort. You must be always fully aware of your environment and must be able in total control of your Beast.
- When driving under wet and slippery conditions, change your behavior accordingly. Riding at a slower speed and braking earlier and gradually are highly recommended as braking distances increase significantly under these conditions.
- Ensure that you are driving at the right speed depending on the terrain and how comfortable you feel.



Please be aware that while driving a S-Pedelec, you could be riding at a considerably faster speed than other bicycles around you. This might be misjudged by others so please drive in accordance with others. More information can be found in the “Appropriate use” section of this manual. Contact us if you are unsure about the appropriate use of this beast and the limits that must be followed.

Instructions about electrics and electronics

! You can find more information about the maintenance, operations, and technical data further in this user manual or in each of the components’ manufacturer’s websites.



The electrical components of a S-Pedelec are more powerful than components from other electric bicycles. Correct use of the beast require regular maintenance checkups from our local partners. They will ensure that all electrics are up to date and are functioning properly which could affect the safety and comfort of your rides.

Immediately turn off the electrical system of the beast and remove the battery if you notice any damage to the electrical components or if parts become exposed after a fall or accident.

Please make sure that you always contact us or our local partners if your Beast needs any repairs, have any technical questions or if you experience and problems of any kind. Insufficient knowledge of the parts can lead to severe injuries or damage!



Always verify that all electrical components have been switched off, that the battery has been removed, and that it has been disconnected from the charger before doing any kind of work or maintenance to your beast.

! If the electric motor does not offer any support, please verify that both brakes are in their original place as their sensors stop the motor from turning on if activated.

General notes about riding

Before riding your new Cargo Beast, please ensure that you have read the following instructions to prevent any possible accidents. Please also consider the operating instructions of each individual components which can be found on the each's manufacturer's website.

If you have any more questions, please contact us.

Before the first ride

Before enjoying your new Beast for the first time, please check the following:

- That all components have been assembled correctly
- That the batteries are fully charged
- That you are familiar with the display's functionalities

! Familiarize yourself with your new beast in a safe and protected area.



New modern brakes may be significantly stronger and may have different braking effects than brakes you are used to. Make sure that you are familiar with them before starting your first ride!

Additionally, remember that brakes can dangerously decrease in effectiveness in wet conditions or on slippery surfaces. Please consider braking softer and for longer distances in order to prevent any accidents.

Before each ride

Before starting every ride, please verify:

- That the seat post is tightly secured and at the correct height
- That the handlebar is attached at the wanted position
- That the tires have the correct air pressure
- That the brakes are working correctly
- That the batteries are properly inserted
- That the transmission is in good condition
- That all the electrical components are safely plugged in

Child seats, child trailers and load trailers



It is not permitted for a child under the prescribed age or for one without the appropriate driving permit to drive a S-Pedelec. If they do, you must thoroughly instruct them on the proper use of the beast before letting them ride unattended. Make sure to explain the dangers of such a powerful electric device.

i Make sure to educate yourself with your nation's regulations before installing anything for your children or a load trailer. These may concern the permitted maximum speed and the total weight. The use of a child trailer on an S-Pedelec is not allowed in many European countries.

As the driver of the beast, you are responsible for taking care of the child and their safety. You must ensure that the child has been thoroughly attached before every ride with one. Driving may also feel different so get used to the sensation on a safe road.

Luggage and load transport

Before every ride with luggage, please verify that nothing can get caught in the spokes and turning wheels.



It is crucial that you always verify that any attached accessories are installed complying with local regulations and the instructions from the manufacturer. Only the use of add-on parts that comply with the respective legal and road traffic regulations is authorized. Using unlicensed parts may lead to accidents so it is recommended that you only use original accessories and add-on parts that fit your Beast. Contact us or visit our website for more information.

Front-wheel luggage racks may not be installed or operated for safety purposes.



Luggage loads are known to change the driving behavior of your Beast. Expect to brake earlier than usual and a more sluggish steering experience. It is also important to note to only transport luggage on the luggage rack intended for such purposes. Incorrect use could lead to severe accidents or falls.

! When riding with luggage, it is important that to verify that the maximum permissible weight of your Beast has not been exceeded. Information on the weight capacity of the rack is also stated in this manual. You can also find more information by contacting our local partners.

Legal requirements

! Electric micro-mobility regulations and requirements are constantly being redefined and may change over time. Please ensure that you are up to date with the legal requirements on your specific type of electric beast. Additionally, every country has their own legislations, so it is recommended that you prepare in advance your trip to avoid any legal issues.

For S-Pedelecs, special laws apply due to their powerful components in order to limit their use. Depending on the country and local government, this type of beast can be considered an electric bicycle or a moped.

For more information, you can either contact us or visit local bike and e-bike associations. Information like what needs to be equipped and driving regulations on public roads can be found and will be useful for traveling. Helmet requirements will also be available in such places. Due to the additional power, S-Pedelecs are also usually have age restrictions and a specific driving license is needed. Laws for children riding on public roads can also be found in such places.

! If you have a private third party-liability insurance, please verify if it covers possible damage caused by your Cargo Beast.

Appropriate use



The Cargo Beast is meant to be an electric bicycle ridden by one person. Depending on the country in which you are riding, you may or may not be allowed to transport children in trailers of S-Pedelec. Additionally, this is a powerful electric bicycle, so we advise to be careful when transporting other people or when you feel uncomfortable while driving.

When driving with luggage, it is required that you verify that your beast is equipped with the proper equipment. Thankfully, your Cargo Beast is already equipped with a sturdy luggage rack that can handle heavy loads. Just verify the total weight and that you are comfortable driving with it. A higher center of gravity might lead to a different driving experience which and so accidents if loss of control.

! The total load of the Cargo Beast is 150 kg. The total load weight allowed: Weight of the Beast including its battery + weight of the driver + weight of total luggage and that of a trailer. Weight of the Cargo Beast is about 38 kg.



Dangers of improper use

Only use your S-Pedelec for its intended purpose and use. This requires a commitment to the operating, servicing and maintenance conditions and information that are found in this manual. If someone else is planning on using your beast, inform them as well of the proper usage and the dangers that can come if not respecting it.

Improper use and non-compliance of the rules and maintenance of your Beast could lead to falls and accidents that could be followed by severe injuries to yourself and others around you.

Cargo Beast can be ridden on public roads as long as it is equipped with the proper equipment defined by law.

Miloo is not liable for damages caused by improper use of the beast. This especially applies to non-compliance of safety instructions such as use on roads, negligence of defects or overloading.

Replacing parts on your Beast

In order to replace specific parts of your Beast, please verify that the replacing parts have been permitted to be used with S-Pedelec class bicycles and with your specific beast in particular.

If you would like to replace components yourself, only original or approved replacement parts are authorized to be used. If not, you will have to seek permission from your local regulatory authority.

Parts that must only be replaced with approved or like-for-like parts are the following:

- | | | | |
|--------------------|--------------|----------------|-----------------------|
| - Battery | - Frame | - Brake system | - Rims |
| - Electronic unit | - Fork | - Seat post | - Tires |
| - Front light | - Stem | - Pedals | - Number plate holder |
| - Rear/brake light | - Handlebars | - Side stand | - Hubs |

Operating elements

In this segment, you will find information on the electric controls of your Beast and instructions on how to use it. Such functions can be used safely while riding by not needing to take your hands off the handlebar.

! If you would like some more information, please check on each manufacturer's website or contact us.

Power Button

The power button of the Cargo Beast is located on the right-hand side of the bike frame. This is the only button to not be situated on the handlebar. With it, you can turn fully turn on or off the electric components of the beast.

İf you would like some more information, please check on each manufacturer's website or contact us.

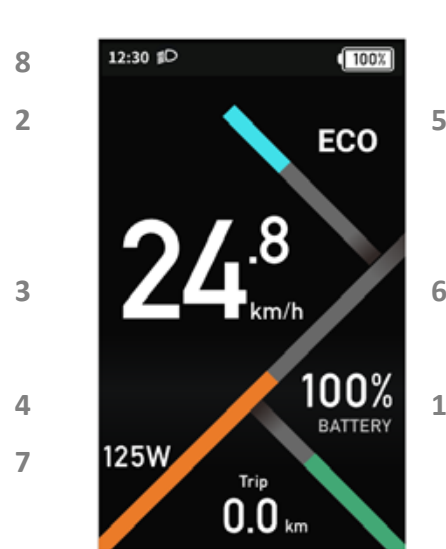
Display

On every Beast, there is an electronics information display unit with that presents the information on your battery level, if you are connected via Bluetooth with your phone, and the speed at which you are riding. You can control the electric support level of the electric motor.

İ After powering up the Cargo Beast, the display needs to be activated separately to be able to use the different electric support levels.

Information:

Once powered on, the display will present multiple information.



1. Battery capacity indication
2. Warn code indication
3. Speed in real-time
4. Speed bar
5. Power-assisted mode indication (4 Modes/ 6 Modes)
6. Unit switching of speed (km/h, mph)
7. Multifunction indication (Clock, TRIP, ODO, MAX, AVG, Range, CAL, Cadence, Time)
8. Symbol indication (Headlight, USB, Service, Bluetooth)



Support Levels:

The + and – buttons of the HMI control make it possible to change between the multiple support modes. The lowest level is Eco and can go up to Boost (5 gears).

! When the system is switched on, the support level will automatically start at 0, for safety purpose.

The supported levels are (in km/h):	Corresponding range (in km):
Off No support level	- N/A
Eco Up to 25	- Up to 80
Tour Up to 30	- Up to 70
Sport Up to 45	- Up to 60
Sport+ Up to 45	- Up to 50
Boost Up to 45	- Up to 40

! Range can drastically vary depending on many factors including terrain, incline and temperature. Please ensure that you prepared for your trip accordingly.

Type-C charging socket:

While riding your Beast, your phone's battery level might cause issues for other cyclists, but not for you. Every Miloo Beast is equipped with a Type-C charging socket behind the display. Just remove the rubber lip and plug in your smartphone to recharge the battery and enjoy a carefree ride.

IOT

This is the optional device for every Beast, it is an internet of things module unit with that presents all information of your bike, which contain multi-module GPS positioning, BLE5.2 communication, 4G communication, beep, and OTA functions.

At first using connect it by Bluetooth with your phone, after that remotely control your bike get possible, by 4G connection you can remotely check the status & location of the bike, lock and release, alarm, and find my bike via beep sound.

When your cell phone close with IOT, it's connected automatically by BLE, at this circumstance you can control the electric support level of the electric motor, personalize each of your power assistance level, complete OTA when needed.

Now let's ready for use,

1. Download the app Bafang Go+ from app store Android & Apple.
2. Register via your personal Email address on app.
3. Try and enjoy this new interface setup on your phone, all this adjustment will synchronize with the bike display.
4. If you need some technical help, please contact us for further support.

Throttle

Placed on the left-hand side of the handlebar, you will find a throttle. This can be used to immediately activate the electric support while riding without needing to pedal.

You can also achieve different speeds depending on how far you press the throttle. This can be used to safely slow down without needing to change the support level.



Depending on the support level, the electric boost can surprise when activated so please ensure that you are comfortable when riding. Before pressing on the throttle, please make sure that both brakes are in their original position as their sensors will stop the electric motor from offering support if activated.

Controller unit (Lamps Indicator)

These buttons are the main controls of the essential features of the Cargo Beast. All these features can be used while riding without any electric support as long as the main button is switched on.

These buttons are all part of the same unit situated on the left hand side of the handlebar, left of the throttle.

Lighting system:

On every Cargo Beast there are integrated lights positioned at different places in order to increase safety while traveling.



While cycling, you can decide to use the front light on high or low beam mode. This can be switched thanks to the upper button of the controller unit on the handlebar. Low beam mode is dedicated to be used while driving through urban areas to increase visibility and let others detect you faster. High beam mode is safer to use in poorly lit areas or while driving through nature in order to get a greater view of your surroundings.

! When your Beast is turned on, the front light will automatically go to low beam mode and cannot be switched off for safety reasons.

Additionally, there is also a rear light that is connected to the brakes and emits a more powerful light when braking.

! Once the S-Pedelec is on, the back light will always emit light and cannot be switched off.



On the controller unit, below the front light, you will find the controls for the turning signals. These will activate the front and rear blinkers to safely warn other around you in which direction you are turning without taking your hands off the handlebar.

Besides you will get the turning signal active reminder with a buzzer notice.

! After turning, the button needs to be pressed again to deactivate the turning signal.

Electronic bike horn:



Your Cargo Beast is supplied with an electronic bike horn to be used to create call attention to yourself in case of dangerous situations. The button is situated at the bottom of the controller unit.

Handling the battery

! The battery is considered to be a hazardous component. Please remove the battery before doing any work to the beast, such as maintenance or assembly. Unintentionally activating the beast could lead to injuries or shock.

Please ensure that you have fully read this section before handling the battery.

Battery safety notes

Before manipulating the battery, please make sure that you are comfortable with all of these warnings.

- Only use the charger that has been given to you with the Cargo Beast. The use of other chargers has a fire risk.
- Do not drop or throw the battery and avoid any strong impact. This could lead to fluid leak, fire or even an explosion.
- Do not apply any force of any kind to the battery. The integrated protection mechanism may get damaged by this deformation and could lead to a fire or explosion.
- Under any circumstance should you use the battery when it is damaged. The liquid inside might leak and any contact with the eye could cause blindness. Avoid any contact, or rinse off the liquid with a large amount of water and we also recommend contacting your doctor.
- Don't ever to attempt to open the battery as it could lead to a short circuit. Opening the battery will void any warranty and guarantee claims.
- Do not store the battery with metal objects that may cause short circuit.
- It is highly recommended to not store multiple batteries on top or next to each other. This could create a chain reaction if something wrong happens to one battery.
- Keep the battery away from any type of heat.
- Protect the battery from encountering water or other type of fluids. This could damage the protective circuit and the protective mechanism of the battery which could lead to explosions.
- Do not clean the battery with a pressure washer. It is recommended to use a damp rag to do so. Avoid at all cost corrosive cleaning solutions.
- In the event of fumes after inappropriate use damage, ensure that the battery is exposed to fresh air. Consider contacting your doctor if you experience any issues.
- Avoid deep discharge of the battery as it may result in irreversible damage to the cells.

- Before any ride, ensure that the battery has been completely attached to the beast and that it cannot fall off.
- The battery is solely intended to be used with the Cargo Beast. Inappropriate use or handling could lead to an injury or a fire. Miloo is not liable for damage caused by bad use.
- For transportation, remove the battery of beast and transport it separately.
- A battery should never be sent by mail. As batteries are considered to be hazardous goods, they should only be sent via your Miloo dealer or by our local partners. This is to avoid overheating and catching fire which could happen under certain circumstances.
- Search information regarding the current regulations about transportation of the Beast.
- Battery cannot be disposed of alongside domestic waste. Either dispose of it at your local recycling dump, at your Miloo dealer or with our partners.



The battery is considered to be a hazardous good, so it is important to know how to take care of the battery in case of puncture or leak. Always wear protective gear when handling a damaged battery, cover all ignition sources with dry non-inflammable materials, move the battery to a well ventilated area to prevent gas accumulation, and avoid any release to the environment such as waterways, sewers, basements or confined areas.

Serious harm can come from mishandling the battery so it is highly recommended to contact a professional before further contact with the damaged battery.

Storage

It is important to store the battery in a dry and well-ventilated room. Keeping the battery at room temperature is optimal to prevent further loss of capacity. Ensure that your Beast is powered off when keep the battery stored in the beast and check the charging level every month or two and recharge it when necessary.

During long periods of storage, the battery should be stored with a lowest charging state of 40%. This to the best way to prevent deep discharging. Verify that the battery is never stored at 0% capacity as it could lead to deep discharging and so irreversible damage to the cells. Possible guarantee or warranty claims could then become void. In order to prevent deep discharging, the battery should be recharged with a span of 2 or 3 days after being fully discharged.

! In order to best preserve your battery, a level range of 40% to 70% is best suited for storage. It is important to take note that this does shorten the period after which a battery can reach deep discharge compared to a fully charged battery. Additionally, it is important to understand that batteries decrease in capacity with age, even when it is well-maintained.

Consider changing a battery that has not been used or charged properly after long periods of time before using your Beast.



It recommended to store the battery at a temperature range of -20 to 60 Celsius. Discharging should be done in the same range but it is recommended to charge your battery between 0 and 45 Celsius.

Removing / Inserting

! Only remove or insert a battery while in a stationary position.

To remove a battery, please follow these quick steps:

Main frame battery:

- Insert the key in its slot on the left hand side of the main frame of the Beast and turn it anticlockwise
- Turn the battery switch below the main frame
- Pull the battery out

To insert a battery, please follow these quick steps:

Main frame battery:

- Slide the battery back in pins facing downwards

! After locking the battery, ensure that it does not move and cannot be removed.



Batteries are heavy goods so ensure that you do not drop them. This could lead to injuries or damaging the battery.

Safety information regarding the charger

! It is important that you read the operating instructions for the charger before using it.

You have two options to charge your Cargo Beast, you can either plug in the charger directly into the beast or remove the battery and charge it separately. It is recommended to charge it in a warmer environment when it is cool outside as it will shorten the time needed to charge. Charging in an area at a lower temperature than 0, higher than 45 or in a tightly sealed space is highly unrecommended.

Be aware that condensation can build up in the charger if there is a sudden change of temperature from cold to warm. This can be avoided by storing the battery in the same place as where it will be charged. It is also important that only the charger given with the Beast is used.

Only use it in a dry room and it is important not to cover it when charging as it could potentially cause a short circuit or fire.

It is also important to verify the mains voltage before plugging in. The voltage written on the type of plate of the charger must correspond with that of the power source.

Always unplug the charger from the socket before cleaning it and it is beneficial for the battery if the charger is disconnected once the charging process has been completed.



It is important that the charger is always placed on a dry and non-flammable surface.



A broken battery should not be used under any circumstance. Additionally, a battery can warm up while charging and immediately unplug if it gets strongly hot.



Do not ride your Cargo Beast when the battery is not inserted. The cover is directly attached to the battery and its absence will not protect from water damage which could ruin the electric components of the beast.

Charging

! Take note of the charger's information first.

In order to charge your battery, you have two options:

1. You can charge the beast by peeling off the rubber seal from the socket placed on the left hand side of the frame and connecting the charger.
2. You can charge the batteries separately from the socket placed at the bottom of the batteries and plugging them in.

! The LED on the charger will light up either in red to express that the battery is still charging, or in green to say that it has finished charging.

Charging time may vary depending on the state of the battery and that of the charger. Charging should take approximately 6 hours to go from 0 to 100%. If the charging time surpasses that written by a considerable amount, we recommend to immediately abort the charging process and change the charger.

Once the charge has finished, it is important to disconnect the battery from the charger, then to unplug the charger from the power source.

! More information on charging is available by contacting us or our local partners.

Service/Maintenance

! Regularly have your electric bike inspected by our local partners. As experts, they can identify if any components are damaged or worn out and can suggest parts that could replace that ones you have. They can also verify the current software used and can update it if necessary. It is recommended not to replace or fix key components of your beast (frame, fork, handlebar, stem, brakes, lights).

! Your S-Pedelec can only function properly, safely, and effectively as long as you replace parts with the relevant and authorized replacement parts. Please contact our local partners for consultations about specific parts.

! You should only replace broken- or worn-down parts with original replacement parts or parts approved by the original manufacturer. These components have been listed in the “Replacing parts on your Beast” segment. In the case of S-Pedelecs, using non approved parts will invalidate the registration of the beast. The use of such parts will also invalidate the warranty and guarantees with the manufacturer for other parts.



Installing an incorrect replacement part can lead to severe loss of function or control of the beast. Components that could have the most severe of consequences are tires with bad grips or that were not designed for the beast, brake pads with inappropriate friction coefficient and wrongly used or badly designed light weight components. Improper assembly could also create the same consequences.

Transmission

SRAM 8 Gears (By default)

Our Cargo Beast is equipped with a SRAM 8-speed transmission system, which is user-friendly and suitable for daily commuting. The gear shifter is mounted on the right side of the handlebar, equipped with a gear indicator window. Please regularly inspect the chain, sprocket, crankset and derailleur for wear and tear.

Rims / Tires



Rims are some of the components that are subject to the greatest amount of strain and are vital in the proper working of the beast. Do not ride your Beast if it has a rim that has been damaged and have it checked out by our local partners so that they can assess the damage and replace it if necessary. Wear can weaken rims and so can lead to falls and serious accidents if unchecked.



It is also important to regularly verify the tires of your S-Pedelec. The range of allowed pressure can be found on the side of the tires. It is important that this range is respected as they could detach from the rim of the wheel or even burst if not respected. If differing pressures are specified for the rims and the tires, the lower maximum pressure and high minimum pressure should be taken into account.

! The Cargo Beast's tires have a recommended pressure on each of tires' mark.

İ Tires are also subject to wear and tear. Therefore, it is important to regularly check the pressure, tread, and overall condition of the tires. Not every tire is designed for the same types of use and make sure that you pick the correct one. Tires of a S-Pedelec can only be exchanged for authorized and approved tires from the manufacturer. It is also crucial that the correct size is respected. If you require more information, contact our local partners or electric bicycle professionals to ensure the correct tire is used.

İ There are also special tires that can be used for winter. Contact our local partners for more information.

Treating punctures:

İ Please contact our local partners or an authorized S-Pedelec workshop to replace flat tires.

Suspension

The Cargo Beast is designed for heavy-duty loading. It is not equipped with suspension components to ensure more stable steering performance. If you have any specific suspension requirements, please consult our in-shop mechanic for further assistance.

Brakes

Your Cargo Beast has been equipped with a pair of hydraulic brakes with integrated electric stop sensors.



Brakes are vital to the good use of and safety on the e-bike. Therefore it is very important to regularly maintain them. This does require specialist knowledge and specific tools so it is recommended that you contact our local partners so that they can do the entire replacement for you. Improper work on the brakes can lead to accidents and so possible injuries.

İ Please note that disc brakes and pads are subject to heavy wear. It is recommended to have a specialist s-pedelec workshop to check these essential components on a regular basis and replace anything worn if needed.



Do not touch the brake disc while it is rotating or directly after braking. This could result in injuries or burn.

Safety instructions for hydraulic disk brakes:



It is recommended to not permanently brake for long periods (for example during long and steep descents). This could create vapor bubbles and cause complete failure of the braking mechanism, which could cause serious accidents and injuries.

Additionally, it should not be activated while the Beast is on its side or upside down. This could allow for air bubbles to enter the hydraulic mechanism which could also lead to failure in the system. After transportation, verify if the pressure point of the brake is still functioning the same as before. If you feel any change, repeatedly apply the brakes slowly in order for the mechanism to get rid of the bubbles.

If the system still doesn't work the same, please contact our local partner in order to fix the issue.

! In order to prevent such bubbles during transport, activate the lever and fix the brake in position with a strap. This would prevent any air to enter the hydraulic system.

! When it is time to clean the hydraulic brakes, it is recommended that you thoroughly read the instructions offered by the component manufacturer.

You can also get more information about the braking mechanism on this S-Pedelec, as well as cleaning and maintenance in the manufacturer's manual or website.

Gears



Gears are considered a key component of the beast and so for your safety while riding. Consider letting our local partner to take on possible work to the gears of your Beast. Improper work that has been carried out can lead high chance of endangering yourself while riding.

! For more information on the gears or maintenance or cleaning of them, please check the manufacturer's manual or website.

Tuning



It is prohibited to apply any modification to the existing technology of your S-Pedelec. Increasing performance or higher speeds could result in legal issues and could make the beast less safe to ride.

Inspections

Modern S-Pedelec technology provides high performance but need to be regularly maintained by qualified people. This work needs specialized knowledge and specific tools, so it is recommended to leave the work to professionals.

To ensure that your Beast remains in a safe condition and fulfills the condition of the warranty, the following terms apply:

- Clean your electric bicycle after every ride and check for damages.
- Have inspections carried out by our local partners.
- Check your Cargo Beast every 100 – 300 km or every three to six months.
- **Verify that all screws, nuts and quick release axles are secure.**
- Use a torque wrench when tightening screw joints.
- Clean and grease any moving parts (except for brake brakes) according to the manufacturer's instructions.
- Have the paintwork touched up.
- Ask a specialist retailer for replacements of any worn or broken parts.

For more information on any parts or maintenance or cleaning of them, please check the manufacturer's manual or website.

For all the plastics parts, i.e. HMI control please make sure it's in right position before tightening it, and the recommend torque is less than 1NM.

Inspection work:

Check the following:

- Tires and wheels

Torques:

- Handlebars
- Cranks

Make adjustments to the following components:

- Brakes
- Pedals
- All mounting screws
- Seat post
- Seat
- Gear system
- Suspension elements

Before each use of your Beast:

Check the following:

- Spokes
- Rims for wear and concentricity
- Tires for damage and foreign bodies and correct air pressure
- Quick release axles
- The functionality of the gear system
- The functionality of the brakes
- Hydraulic brake: tightness
- **Rear hub screws on position**
- Lighting
- Horn
- Tires: safe seat and correct air pressure

Every 100 – 300 km:

Check the following:

- Drive chain
- Check the brake discs for wear, replace them if required
- All screw joints are secure

Clean:

- Drive chain

Every 1000 km:

To be checked, cleaned or possibly replaced by our local partners:

- Hubs
- Pedals
- Brakes
- Electric motor
- Battery
- Gear system

After rides in the rain:

Work to be undertaken:

- Remove humidity, dry

Clean:

- Gears

! The first inspection of the beast is very important to ensure its correct use and safety. Cables can stretch and bolted connections can loosen. Therefore, please leave the first inspection to our local partners.

Warranty, defects, wear

When purchasing your Cargo Beast, you should have received the warranty and limitation of liability separately from your beast. For more information, please contact us.

Please note that the components of a S-Pedelec are subject to higher wear than components from other bicycles. This is due to higher average speed of the beast that the motor can achieve and the superior weight of the components.

This wear is therefore not a defect and is not subject to warranty.

Components that are mostly affected are the tires, brake pads and drive components.

The battery is also subject to aging and so a wearing part as well. Its age and duration of use will affect its effectiveness. Please consider changing your battery for a new one after a few years, and so contact us for more information.



The Beast bike has been designed to be used in general conditions. Don't jump with your bike. Jumping a bike, particularly the mountain bike, can be fun, but it can put huge and unpredictable stress on the bicycle and its components. Riders who insist on jumping their bikes risk serious damage, to their bicycles as well as to themselves.

Technical data

Legal total weight of Cargo Beast: 146 kg
Beast including driver, battery, luggage, trailer
Cargo Beast Weight: 38 kg

Motor	Bafang RM G0630.750 DC
- Motor Support	Up to 45 km/h
- Power [W]	750
- Range [km]	Up to 80
Li-ion Batteries	LG
- Voltage [V]	48
- Capacity [Ah]	14
- Energy [Wh]	672
- Charging Time [h:min]	8:00

Charger	SANZ
- Input	AC 100-240 V / 47-63 Hz / 3.5 A
- Output	54.6 V / 3 A
- Operating temperature range [°C]	0 -> 40
- Permitted storage temperature range [°C]	-20 -> 60
- Permitted discharging temperature range [°C]	-20 -> 60
- Permitted charging temperature range [°C]	0 -> 45
- Approximate weight [kg]	4

! If you see that the charging time is exceeding the indicated value, please consider stopping and have the battery and charger checked by our local partners.

Environmental tips

While dealing with packaging materials, dispose of them according to your local waste management system.

General care and cleaning products

When taking care or cleaning your S-Pedelec, take the environment into consideration. You should only use products that are biodegradable when possible.

Brake cleaner and lubricants

Please ensure that you use a suitable brake cleaning tool and dispose of brake fluids properly at a suitable waste disposal site. Lubricants should also be treated with the same amount of care.

Tires and inner tubes

Tires and inner tubes are not residual waste and do not belong in domestic rubbish. They have to be disposed at your local recycling center.

Batteries

S-Pedelecs have batteries that are considered to be hazardous and so are subject to compulsory labeling. They should be disposed at Miloo centers or by the manufacturers.

Legal disclosure

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Version 1.0

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